

Rishi Raj

Doctoral Researcher in Mathematical Physics • Aspiring Machine Learning Engineer

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Versatile and results-driven Machine Learning Engineer with a strong foundation in web application development and applied machine learning. Proficient in building scalable AI-powered applications using modern frameworks like Node.js, React, and TypeScript, and integrating robust machine learning workflows with Python.

Expert in deploying and optimizing AI solutions, with hands-on experience in large language models (LLMs), natural language processing (NLP), and cloud-based model deployments. Skilled in MLOps/DevOps best practices, ensuring efficient, reliable, and high-performance AI pipelines.

Collaborative and innovation-focused professional, adept at working with cross-functional teams to solve real-world challenges. Committed to staying updated with advancements in AI and applying cutting-edge technologies to deliver transformative solutions.

TECHNICAL SKILLS

Languages: Proficient in Python (scientific and production-grade), C/C++ (high-performance computing), JavaScript / TypeScript (full-stack systems)

ML & AI Frameworks: PyTorch, scikit-learn, XGBoost, Hugging Face, Transformers, Weights & Biases (wandb)

Cloud & DevOps: End-to-end model deployment pipelines with AWS & GCP, Docker, CI/CD via GitHub Actions, ML model deployment & monitoring, MLOps best practices

Scientific Computing: Mathematica, GNU Scientific Library (GSL), Boost C++, Julia, SageMath

Dev Tools & Version Control: Advanced Git workflows, Linux System Administration, VS Code

Full Stack Web Development: Node.js, Flask, FastAPI, React based frameworks like Next.js, MongoDB, SQL

Core Competencies: Research-grade problem solving, Continuous Learning, Collaborative Problem-Solving, Cross-Functional Communication, Leadership, Adaptability

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

May 2023 – Jul 2023

Supervised by Boris Pioline @ LPTHE, Sorbonne Université, Paris, France

- Utilized Mathematica and SageMath to study bound states of BPS particles in the Coulomb branch of $\mathcal{N} = 4$ supersymmetric quiver quantum mechanics (QQM), revealing structural features of the moduli space.
- Performed a comparative analysis of BPS index computation via attractor flow trees, Coulomb branch localization, and modular anomaly methods, offering insights into wall-crossing behavior.
- Helped clarify the contribution of scaling solutions using the Minimum Modification Hypothesis (MMH), reproducing attractor flow tree results from the Coulomb branch formula.
- Applied graph-theoretic and numerical methods to compute physical invariants, improving precision and performance in multi-centered black hole configurations.

Web Developer Intern

Nov 2021 – Jul 2022

Zenoholics Pvt. Ltd., Bangalore, India

- Spearheaded modernization of full-stack applications using React.js, Next.js, and MongoDB, achieving a 30% performance gain through architectural refinement and performance profiling.
- Leveraged advanced diagnostics tools—DevTools, Profiler API, Lighthouse, Postman, and Bundle Analyzer—to identify bottlenecks and guide optimization across the stack.
- Engineered robust CI/CD workflows with GitHub Actions, Docker, and AWS; incorporated Kubernetes for container orchestration and scalable deployments.
- Gained hands-on experience in scaling distributed systems via RabbitMQ, connection pooling, caching strategies, lazy loading, CDNs, and load testing, enhancing backend resilience and frontend responsiveness.

EDUCATION

Doctor of Philosophy, PhD in Mathematical Physics

Oct 2023 – Sept 2026 (expected)

Sorbonne Université, Paris, France

TODO Research focus on *BPS Black Holes and Donaldson-Thomas Invariants* under the supervision of Boris Pioline. Currently exploring phase space partitioning and attractor flow trees.

Advanced Deep Learning (ongoing), Applied Machine Learning, String Theory, Quantum Field Theory II, Supersymmetry and Supergravity, Differential Geometry

Master of Science, MS in Physics

Aug 2022 – Jul 2023

Indian Institute of Technology, IIT Madras, Chennai, India

9.18/10 CGPA

Received the *Mr S. Venkitaramanan, I.A.S Retd* prize and *Electronics For You* prize for securing the highest CGPA during two academic years. Master's thesis on *Black Holes through the lens of Matrix Theory*.

Quantum Field Theory, Advanced General Relativity, Statistical Physics, Numerical Methods and Programming Lab, Conformal Field Theory, Advanced Quantum Computation and Quantum Information

Bachelor of Science, BS in Physics

July 2018 – Aug 2022

Indian Institute of Technology, IIT Madras, Chennai, India

9.18/10 CGPA

Graduated with a Minor in Mathematics and consistently maintained a high CGPA. Conducted significant research in theoretical physics through multiple projects and publications.

Mathematical Physics, Dynamical Systems and Chaos, Differential Topology, Real Analysis, Algebra I, Quantum Mechanics, Probability and Stochastic Processes, Functional Analysis

ACADEMIC PROJECTS

Machine Learning Bulk Reconstruction

Jan 2023 – Apr 2023

Developed neural networks to reconstruct bulk geometry from boundary physics in holographic setups.

- Trained models on numerical relativity data for asymptotically $\text{AdS}_4/\text{CFT}_3$ systems, demonstrating high-accuracy reconstruction.
- Validated network performance rigorously and analyzed deviations from Einstein's equations to ensure reliability.

Matrix Models and Black Holes

Dec 2021 – Dec 2022

Designed a simulation suite for classical and quantum simulations of the M-theory matrix model.

- Achieved performance improvements of up to 17x for simulations of black hole dynamics described by random matrix ensembles.
- Studied deviations in thermalization behavior to identify underlying quantum mechanical properties.

Magnetically Coupled Oscillators

Nov 2019 – Jan 2020

Created an experimental setup and theoretical model for magnetically coupled pendula in a research fellowship.

- Designed and conducted experiments to capture data on coupled oscillators with magnetic interactions.
- Built a theoretical framework and performed comparisons between experimental and simulated results, ensuring consistency.

Semiholographic Networks

Jul 2019 – Nov 2020

Developed scalar field networks coupled with perfect fluids using a semiholographic framework.

- Implemented the theoretical models and analyzed perturbation responses to understand dynamical properties.
- Collaborated with supervisors to produce insights on emergent behaviors in coupled systems.

arXiv-Rename and PIRSA-DL Automation Tools

Jan 2021 – Jun 2022

Engineered Python CLI tools for automating workflows in research and lecture management.

- Developed **arXiv-Rename**, a tool that uses arXiv APIs to rename files based on author and title metadata.
- Built **PIRSA-DL**, a utility for scraping and downloading lecture content from the Perimeter Institute archive.

OTHER WORK EXPERIENCE

Teaching Assistant

Jul 2022 – Nov 2022

PH5060 (Physics Lab I), IIT Madras, Chennai, India

- Assisted students with computational physics problems, including chaotic systems and Monte Carlo simulations.
- Conducted viva sessions and evaluated reports to assess students' understanding and progress.

ACTIVITIES

Horizon Club, IIT Madras

Sep 2019

Delivered a lecture series titled *Relativity from Symmetries*, introducing group theory concepts and special relativity to physics enthusiasts

- Designed and presented four lectures focusing on symmetry principles, Lorentz transformations, and physical implications of special relativity.
- Engaged and inspired beginning physics enthusiasts to explore advanced theoretical concepts through interactive discussions.

Music Club, IIT Madras

Jul 2019 – Nov 2019

Organized and coordinated musical events, including performances for the annual cultural festival, Saarang

- Led the planning and execution of multiple musical events, ensuring seamless coordination among performers and technical teams.
- Enhanced event management and teamwork skills while handling responsibilities for the institute's cultural programs.